POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Fall

Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define buffer solution Explain the mechanism of acidic buffer solution. Calculate the pH of a buffer solution prepared by mixing 400 ml of 0.5M sodium acetate and 800 ml of 0.1M acetic acid which is 1.3% ionized in dilute solution. 8

   b) What is standard hydrogen electrode? How can we determine the standard electrode potential of zinc electrode by using SHE? Determine the reduction potential of zinc electrode when it is dipped in to 0.1M $ZnSO_4$ solution at 25° C. Given standard oxidation potential of Zinc is 0.76 V. 7

2. a) Explain why ionization energy value of an ion increases as number of positive charge increases. Discuss the factors affecting its value . 7

   b) Give reasons:

      i. Transition elements show variable oxidation states.

      ii. Mn can form complex compounds but not Mg. 8

      iii. Zn is not considered as true transition element.

      iv. Transition elements are mostly paramagnetic.

3. a) Describe the reaction mechanism and stereochemistry of SN^1 reaction taking suitable example. What are the factors governing the rate of SN^1 and SN^2 reaction? 8

   b) Define carbocation. How are carbocations stabilized? Write any two methods of their formation. 7

4. a) Explain the procedures involved in the preparation of rubber. What do you mean by vulcanization of rubber? 7

OR

Point out the major limitations of Bohr's atomic theory. Derive an

expression so as to calculate the radius of Bohr's third orbit of H-atom.

b) Write the preparation, properties and uses of PVC. Write the point of differences between thermosetting & thermoplastic polymer. 8

a) Explain the principle, and procedures involved for the separation of mixtures by thin layer chromatography. 8

OR

Define hybridization. Explain the formation of CH_4 molecule on the basis of hybridization. What is the cause of variation of bond angles between CH_4, H_2O and NH_3 molecules?

b) What are lubricants? How are they classified? Write down their uses. 7

OR

What is metallic bonding? How does electron sea model of metallic bonding explain the metallic properties like metallic luster, thermal and electrical conductivity?

a) What is ozone layer depletion? Mention the main causes of it. Explain the consequences of ozone layer depletion. What should be done to control it? 8

b) What are the causes of hardness of water? How can it be measured in the laboratory? Describe it. 7

Write short notes on: (Any two) 2×5

a) Corrosion, its types and prevention

b) Acid rain

c) Markovnikoff's rule